Hepatocellular carcinoma metastasis to salivary glands: a systematic review of diagnostic challenges and survival outcomes

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Background: Hepatocellular carcinoma (HCC) metastasizes predominantly to the lungs, lymph nodes, and bone; salivary gland involvement is exceedingly rare and poorly characterized. We conducted a systematic review to define the clinical profile, diagnostic challenges, and outcomes of this unusual metastatic pattern.

Methods: Following PRISMA 2020 guidelines with prospective protocol registration (PROSPERO CRD42024559907), we searched six databases from inception through November 2024 for pathologically confirmed HCC salivary gland metastasis. Two reviewers independently screened records, extracted patient-level data, and appraised study quality. Outcomes were summarized using descriptive statistics and Kaplan-Meier survival analysis.

Results: From 148 records, eight published cases met inclusion criteria; combined with one institutional case, the final cohort comprised nine patients [1998–2024]. All were male (median age 64 years; range, 36–82 years). Alcohol use disorder (55.6%) and diabetes mellitus (33.3%) were the most common comorbidities. The parotid gland was affected in 88.9% of cases. Two-thirds of cases presented synchronously with primary HCC; among three metachronous cases, the median interval to salivary metastasis was 2.5 years, including one case arising after liver transplantation. Initial fine-needle aspiration biopsy (FNAB) was diagnostic in only 2 of 8 attempted cases (25%); 4 of 8 ultimately required surgical biopsy. Among seven patients with survival data, median overall survival was 6 months (range, 2–24 months). Patients undergoing surgical resection had numerically longer median survival (12 *vs.* 5 months); no formal statistical comparison was performed given the small sample size.

Conclusions: HCC salivary gland metastasis has been observed exclusively in males across all reported cases, with strong parotid predilection and poor prognosis. FNAB has a low diagnostic yield, and early surgical biopsy should be considered when clinical suspicion is high. Delayed metastases, including post-transplantation, underscore the need for extended surveillance and sustained clinical vigilance in HCC follow-up.

Keywords: Hepatocellular carcinoma (HCC); salivary gland neoplasms; neoplasm metastasis; parotid gland; liver transplantation

Submitted Jan 07, 2026. Accepted for publication Apr 16, 2026. Published online May 18, 2026.
doi: 10.21037/jgo-2026-1-0015
View this article at: https://dx.doi.org/10.21037/jgo-2026-1-0015

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Introduction

Background

Hepatocellular carcinoma (HCC) is the most common primary liver malignancy and the third leading cause of cancer-related mortality worldwide, accounting for approximately 906,000 new cases and 830,000 deaths annually (1,2). Despite advances in surgical resection, liver transplantation, locoregional therapies, and systemic agents, prognosis remains poor; 5-year survival is approximately 18% across all stages and less than 3% for distant metastatic disease (3).

Extrahepatic metastasis complicates 13.5–42% of HCC cases over the course of disease, most frequently involving the lungs (55%), lymph nodes (53%), bone (28%), and adrenal glands (11%) (4). When extrahepatic disease is identified, median survival is measured in months and treatment intent typically shifts toward palliation. Metastasis to unusual anatomical sites remains poorly characterized in the literature, creating a risk of diagnostic delay and suboptimal management when clinicians encounter unfamiliar presentations.

Salivary gland metastases from any distant primary malignancy are exceedingly rare, accounting for less than 10% of all salivary gland malignancies, with most originating from head and neck primaries (5,6). Spread from hepatobiliary cancers is particularly unusual. Among the major salivary glands, the parotid is disproportionately involved in metastatic disease. Unlike the submandibular and sublingual glands, which encapsulate before lymphatic tissue matures, the parotid undergoes delayed encapsulation and incorporates lymph nodes within its parenchyma, creating lymphatic and hematogenous routes of metastatic seeding that are structurally absent in the other major salivary glands (7).

Highlight box

Key findings

- Nine cases of hepatocellular carcinoma (HCC) salivary gland metastasis were identified (1998 to 2024), all in males. The parotid gland was affected in 88.9% of cases.
- Initial fine-needle aspiration biopsy (FNAB) established a definitive HCC diagnosis in only 2 of 8 cases (25%). Of the 6 cases with non-diagnostic cases, 2 were diagnosed on repeat FNAB; 4 required excisional biopsy.
- Median survival after salivary gland diagnosis was 6 months (range, 2–24 months; n=7 with follow-up data). The 1-year survival rate of 42.9% (3/7). Median survival was 12 months in surgical versus 5 months in non-surgical cases (descriptive only; no formal statistical comparison given small sample size).

What is known and what is new?

- HCC most commonly spreads to lungs (55%), lymph nodes (53%), and bone (28%). Extrahepatic metastasis predicts poor prognosis regardless of tumor control.
- All cases involved males; common comorbidities included alcohol use disorder (55.6%) and diabetes (33.3%). Two-thirds of patients presented synchronously with the primary diagnosis. Metachronous cases appeared at a median of 2.5 years. Post-transplant metastasis was documented.

What is the implication, and what should change now?

- HCC can metastasize to the salivary glands years after treatment, including post-transplant. Head-and-neck assessment at routine follow-up and patient education on warning signs may facilitate earlier detection.
- FNAB alone is often insufficient; early surgical biopsy should be considered for suspicious parotid masses when initial FNAB is non-diagnostic.
- Surveillance protocols for transplant recipients may warrant extension beyond standard timeframes.

Rationale and knowledge gap

When an HCC patient presents with a new parotid or submandibular mass, clinicians face a diagnostic and prognostic problem with no systematic evidence base to guide them. The existing literature consists entirely of isolated case reports, leaving the demographic profile, diagnostic performance of available modalities, optimal treatment approach, and expected outcomes undefined. For liver transplant recipients in particular, a population at unique risk due to long-term immunosuppression, it is unclear how surveillance should be adapted to account for late or atypical metastatic recurrence. Without systematic synthesis of available cases, clinicians cannot meaningfully risk-stratify patients, counsel them on prognosis, or make evidence-informed decisions about biopsy strategy.

Objective

This systematic review synthesizes all published evidence on HCC metastasis to salivary glands, with the goals of characterizing patient demographics and comorbidities, describing the timing and anatomical pattern of metastasis, evaluating the diagnostic yield of fine-needle aspiration versus surgical biopsy, summarizing treatment approaches, and describing survival outcomes. We additionally present an institutional case of delayed parotid metastasis arising

2.5 years after liver transplantation, illustrating the diagnostic complexity of this entity in immunosuppressed patients. We present this article in accordance with the PRISMA reporting checklist (8) (available at https://jgo.amegroups.com/article/view/10.21037/jgo-2026-1-0015/rc).

Methods

Protocol and registration

We registered the protocol with the International Prospective Register of Systematic Reviews (PROSPERO; CRD42024559907) prior to data extraction. The inclusion of our institutional case was not pre-specified in the PROSPERO protocol; it was identified as a contemporaneous case at Geisinger Wyoming Valley Medical Center during the conduct of the review and added prospectively prior to data analysis.

Objectives and PICO framework

The review question was structured using the PICO framework as follows. Population (P): adult patients with pathologically confirmed HCC. Index condition (I): occurrence of metastasis to any major salivary gland (parotid, submandibular, or sublingual). Comparator (C): no formal comparator was pre-specified given the descriptive nature of the review; where data permitted, treatment subgroups (surgical resection versus non-operative management) were compared. Outcomes (O): primary outcome was overall survival measured from the date of salivary gland metastasis diagnosis; secondary outcomes included diagnostic modality yield [fine-needle aspiration biopsy (FNAB) versus surgical or core needle biopsy], clinical presentation characteristics, metastasis timing (synchronous versus metachronous), comorbidity profiles, and treatment approaches.

Eligibility criteria

Inclusion criteria

- ❖ Population: we included patients with pathologically confirmed HCC. Eligible patients had documented metastasis to salivary glands, including the parotid, submandibular, or sublingual glands.
- ❖ Study types: we included case reports, case series, and observational studies reporting individual patient-level data.
- ❖ Time period: database inception through November 27, 2024.
- ❖ Language: no language restrictions applied.

Exclusion criteria

- ❖ We excluded studies in which HCC was not the primary tumor or in which other malignancies were present.
- ❖ Lack of pathological confirmation of salivary gland metastasis.
- ❖ Animal or preclinical studies.
- ❖ Conference abstracts, editorials, commentaries, and non-peer-reviewed content.
- ❖ Articles without extractable patient-level clinical data.

Information sources

To ensure our review was as thorough and unbiased as possible, we searched six major electronic databases: PubMed, MEDLINE, EMBASE, Cochrane Library, Web of Science, and Ovid. These searches spanned from each database's inception through November 27, 2024, capturing the maximum possible range of published literature.

Search strategy

We created our search strategy with a medical librarian to increase sensitivity and specificity. We used both Medical Subject Headings (MeSH) and free-text keywords for HCC, such as "HCC", "liver cancer", "liver cell carcinoma", and "hepatoma". We also added terms for metastasis to salivary glands: "metastasis", "metastases", "metastatic", "secondary", "parotid gland", "submandibular gland", "sublingual gland", and "salivary gland". We used Boolean operators to combine and refine results. All searches were last performed on November 27, 2024. The complete search strategies for all six databases, including the full PubMed search string with MeSH terms, Boolean operators, and field tags, are provided in Appendix 1 to allow full reproducibility.

Selection process

Two independent reviewers screened all titles and abstracts identified through the database searches using the Rayyan software for systematic review screening. Prior to screening, duplicate records identified across databases were removed using Rayyan's automated deduplication feature, with manual verification to confirm that no unique records were inadvertently excluded; each patient case was represented only once in the final dataset. Articles deemed potentially eligible by either reviewer underwent full-text evaluation. Both reviewers independently assessed full-text articles against the predefined eligibility criteria. Reviewers settled conflicts about inclusion through discussion. When they could not reach a consensus, a third reviewer adjudicated. We documented the selection process in accordance with the PRISMA 2020 guidelines, with reasons for exclusion recorded at the full-text screening stage.

Data collection process

Two reviewers independently extracted data using a standardized, pilot-tested form designed for this review. A third reviewer checked the extracted data for accuracy and completeness.

Data items

We systematically extracted data from the included studies. We recorded patient demographics: age, sex, race or ethnicity, and geographic location. Next, we documented comorbidities and risk factors. This included the cause of chronic liver disease, such as viral hepatitis, alcohol use disorder, and nonalcoholic steatohepatitis/nonalcoholic fatty liver disease (NASH/NAFLD). We also noted conditions like diabetes, hypertension, and cirrhosis status when available.

We collected primary HCC characteristics. These included date of initial diagnosis, tumor stage, and alpha-fetoprotein (AFP) levels at diagnosis [reported in ng/mL; classified as elevated, normal, or not reported (NR) based on each study's stated reference range or authorial description]. We included treatment types: surgical resection, transplantation, or systemic therapies. For salivary gland metastasis, we recorded location, laterality, timing (synchronous or metachronous), and interval from HCC diagnosis. We also recorded symptoms, physical exam findings, and the presence of other extrahepatic metastases.

In diagnostics, we noted all imaging methods used, including computed tomography (CT), magnetic resonance imaging (MRI), and ultrasound. We evaluated fine-needle aspiration and biopsy results. Treatment strategies included surgical resection, systemic therapies, radiation, and palliative care. Main outcomes were overall survival from diagnosis of salivary metastasis, vital status at last follow-up, and cause of death. We also recorded secondary outcomes on metastasis timing and recurrence patterns.

Study risk of bias assessment

We assessed the methodological quality of all included case reports using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Case Reports (9). This tool comprises eight criteria: (Q1) clear description of patient demographics; (Q2) history presented as a clear timeline; (Q3) clear description of the clinical condition at presentation; (Q4) clear description of diagnostic tests and results; (Q5) clear description of the intervention or treatment; (Q6) clear description of the post-intervention clinical condition; (Q7) identification and description of adverse events; and (Q8) presence of key takeaway lessons. Each criterion was rated as yes (Y), no (N), unclear (U), or not applicable (NA). We classified overall risk of bias as low (≥6 yes responses), moderate (4–5 yes responses), or high (≤3 yes responses). Assessment was performed independently by the first and senior authors; discrepancies were resolved by consensus. Complete results of the JBI appraisal are presented in Appendix 2.

Effect measures

For time-to-event outcomes, the anchor event was the date of pathological confirmation of salivary gland metastasis, defined as the date of the tissue diagnosis that established HCC metastasis, whether by fine-needle aspiration, core needle biopsy, or surgical excision. The endpoint was death from any cause or last known follow-up. We extracted survival times as reported in each source publication. When authors reported survival in months, we used the reported value directly. When survival was described in approximate terms (e.g., "approximately 4 months", "about 6 months"), we used the stated approximate value and noted this in the individual data. When survival data were NR or could not be extracted from the publication (n=2: Dargent *et al.*, 1998; Aiyer *et al.*, 2019), those cases were excluded from the survival analysis and are noted as NR in the outcomes table. Patients confirmed alive at their last documented follow-up visit were censored at that time point. The primary effect measure was overall survival.

Synthesis methods

Given the rarity of this condition and the heterogeneity of the available evidence (predominantly case reports), a meta-analysis was not appropriate. Instead, we employed descriptive synthesis methods to summarize the evidence.

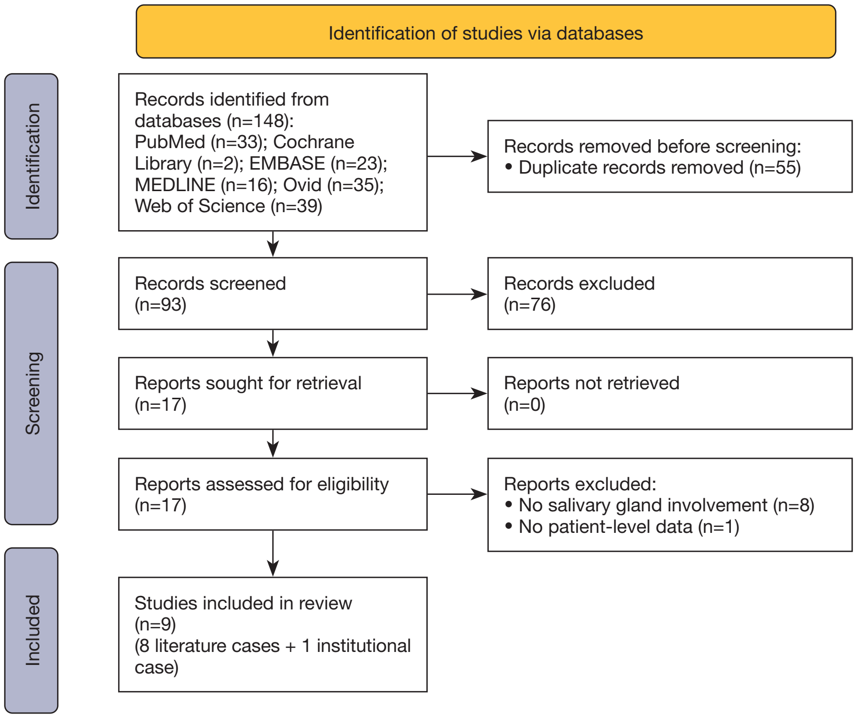

Figure 1 PRISMA flow diagram. Flow diagram showing systematic search and selection process from 148 initial records through database searching to final inclusion of 8 studies meeting criteria, following PRISMA 2020 guidelines.

We report continuous variables as mean ± standard deviation (SD) and median with a range (minimum-maximum). We present categorical variables as frequencies and percentages. For the survival analysis, we reconstructed individual patient-level data from published case reports. We generated Kaplan-Meier survival curves to estimate median overall survival from the date of salivary metastasis diagnosis. Given the small sample size (n=9, with survival data available for 7 patients), survival comparisons between treatment groups are presented as exploratory and descriptive only; no formal hypothesis testing was performed.

We performed all statistical analyses using R version 4.4.0 (R Foundation for Statistical Computing, Vienna, Austria) and the following packages: tidyverse (version 2.0.0) for data manipulation, survival (version 3.5-7) for survival analysis, and survminer (version 0.4.9) for visualizing survival curves.

Given the small sample size, descriptive nature of the evidence, and lack of comparable control groups, we did not perform formal sensitivity analyses. The robustness of our findings is instead assessed through transparent reporting of individual patient-level data and detailed discussion of limitations in the interpretation of results.

Results

Study selection

Our systematic literature search identified 148 records across six electronic databases: PubMed (n=33), Web of Science (n=39), Ovid (n=35), EMBASE (n=23), MEDLINE (n=16), and Cochrane Library (n=2). After removing 55 duplicate records, 93 unique citations underwent title and abstract screening. Of these, 17 articles advanced to full-text review for detailed eligibility assessment. Following full-text evaluation, eight studies met all inclusion criteria and were included in the systematic review. These eight studies reported individual patient-level data for eight patients with HCC salivary gland metastasis. Combined with our institutional case, the final cohort comprised nine patients reported between 1998 and 2024. The study selection process is detailed in the PRISMA flow diagram (*Figure 1*). Among the nine studies excluded at full-text review, the primary reason for exclusion was absence of confirmed salivary gland involvement (n=8); one study was excluded for inability to extract individual patient-level data (n=1).

Table 1 Patient demographics and clinical characteristics

Characteristic	Value
Age, years	
Mean ± SD	61.7±14
Median [IQR]	64 [55–68]
Range	36–82
Male sex, n (%)	9 (100.0)
Comorbidities	
Patients with any comorbidity, n (%)	7 (77.8)
Mean number per patient	2.2
Alcohol use disorder, n/N (%)	5/7 (71.4)
Diabetes mellitus, n (%)	3 (33.3)
Hepatitis C, n (%)	2 (22.2)
Hypertension, n (%)	2 (22.2)
Tobacco use, n (%)	2 (22.2)

IQR, interquartile range; SD, standard deviation.

Study characteristics

The included studies consisted exclusively of case reports (n=8), representing the only available evidence for this rare clinical entity (10-17). Studies originated from diverse geographic regions: United States (n=4, 50%), China (n=1, 12.5%), India (n=1, 12.5%), Italy (n=1, 12.5%), Qatar (n=1, 12.5%), and South Korea (n=1, 12.5%). Publication years ranged from 1998 to 2024, with no temporal clustering observed. All studies provided pathological confirmation of HCC metastasis to the salivary glands via surgical resection or biopsy. We summarized the characteristics of included studies in *Figure 1*.

Risk of bias in studies

JBI Critical Appraisal Checklist scores for all nine case reports are presented in Appendix 2. Six studies (66.7%) were classified as low risk of bias: Romanas *et al.* [2004], Moore *et al.* [2010], Elzouki *et al.* [2014], Yu *et al.* [2013], Deng *et al.* [2021], and the current case. Three studies (33.3%) were classified as moderate risk of bias: Dargent *et al.* [1998], Aiyer *et al.* [2019], and Vitale *et al.* [2009]. No study was classified as high risk. The most common source of methodological uncertainty was incomplete reporting of adverse events (Q7; unclear or NA in 3/9 studies, 33.3%) and post-intervention clinical condition (Q6; unclear or absent in 2/9 studies, 22.2%). The two moderate-risk studies with the lowest scores (Dargent *et al.* and Aiyer *et al.*) each had three criteria rated unclear, primarily reflecting the structural constraints of letter-to-editor and brief report formats rather than true clinical omissions. Overall, the body of evidence was judged to be of acceptable methodological quality given the inherent limitations of single-case reporting.

Results of individual studies

Patient demographics

All nine patients were male (100%), with a mean age of 61.7±14 years at the time of diagnosis of salivary metastasis (median 64 years, range, 36–82 years). Age distribution: 2 patients <50 years (22.2%), five patients aged 50–69 years (55.6%), and two patients ≥70 years (22.2%). The geographic distribution of cases included the United States (n=4, 44.4%), with single cases from China, India, Italy, Qatar, and South Korea, representing diverse international populations (*Table 1*).

Comorbidities and risk factors

Documented comorbidities were present in 7/9 patients (77.8%). Patients had a mean of 2.2 comorbidities per individual (median 2, range, 0–6). The most prevalent comorbidity was alcohol use disorder, present in 5 patients (55.6%), followed by diabetes mellitus in 3 patients (33.3%). The literature documented hepatitis in 3 patients total: hepatitis C (n=2, 22.2%) and hepatitis B (n=1, 11.1%). Cardiovascular comorbidities included hypertension (n=2, 22.2%) and coronary artery disease (n=1, 11.1%). Active tobacco use was reported in 2 patients (22.2%). Additional comorbidities included cirrhosis, hyperlipidemia, and intravenous drug abuse (each n=1, 11.1%). The prevalence of metabolic and substance use comorbidities is consistent with typical HCC risk factor profiles.

Primary HCC characteristics and timing of metastasis

The temporal relationship between HCC diagnosis and salivary metastasis varied substantially. Synchronous presentation, defined as salivary metastasis detected at initial HCC diagnosis or within 1 month, occurred in 6 cases (66.7%). Metachronous presentation, with salivary metastasis developing after initial HCC diagnosis and treatment, occurred in 3 cases (33.3%).

Among the three metachronous cases, the interval from HCC diagnosis to salivary gland metastasis ranged

Table 2 Clinical presentation and anatomical distribution

Characteristic	Value, n (%)
Metastasis site	
Parotid gland	8 (88.9)
Submandibular gland	1 (11.1)
Timing relative to HCC diagnosis	
Synchronous	6 (66.7)
Metachronous	3 (33.3)
Clinical presentation	
Painless swelling	6 (66.7)
Painful/tender mass	3 (33.3)
AFP status at metastasis	
Elevated	3 (33.3)
Normal	3 (33.3)
Not reported	3 (33.3)

AFP, alpha-fetoprotein; HCC, hepatocellular carcinoma.

Table 3 Treatment approaches and survival outcomes

Parameter	Value
Treatment modalities, n (%)	
Surgical resection	3 (33.3)
Superficial parotidectomy	2 (22.2)
Submandibular resection	1 (11.1)
No adjuvant systemic therapy	3 (100.0)
Non-operative management	6 (66.7)
Systemic therapy (chemotherapy and/or hormonal)	4 (44.4)
Sorafenib	2 (22.2)
Combination chemotherapy	1 (11.1)
Hormonal therapy	1 (11.1)
Supportive care only	2 (22.2)
Survival outcomes	
Median overall survival, months	6
Range, months	2–24
Deaths observed, n/N (%)	5/7 (71.4)
Survival by treatment approach	
Surgery: median survival, months	12
No surgery: median survival, months	5

N=7 patients with documented follow-up data. Dargent *et al*. [1998] and Aiyer *et al*. [2019] were excluded from survival analysis due to absence of documented survival time, mortality status, or follow-up data.

from 8 months to 8 years (median 2.5 years). Notably, our institutional transplant case demonstrated a disease-free interval of 2.5 years (30 months) following liver transplantation, with the patient having received bridging transarterial chemoembolization (TACE) therapy before transplant. This case represents the second-longest reported interval between HCC diagnosis and salivary metastasis in the literature.

Clinical presentation

Clinical presentation was generally consistent across cases. The parotid gland was the overwhelmingly predominant site of involvement, affected in 8/9 cases (88.9%), with only 1 case involving the submandibular gland (11.1%). All parotid cases were unilateral; We observed no bilateral involvement (*Table 2*).

The most common presenting symptom was painless swelling or mass, reported in 6 patients (66.7%). Three patients (33.3%) experienced tenderness or pain at the mass site. Additional symptoms included facial numbness (n=1, 11.1%) and systemic symptoms such as fever and unintentional weight loss. Physical examination typically revealed firm, mobile masses without facial nerve involvement. Cervical lymphadenopathy was uncommon.

Two patients (22.2%) presented with concurrent extrahepatic metastases at other sites, including bone and abdominal wall, indicating widespread disseminated disease. However, the majority (7/9, 77.8%) had salivary gland as the sole or presenting site of extrahepatic disease. We described the clinical presentation characteristics in *Table 3*.

Diagnostic approach

The authors presented cross-sectional imaging in 7 patients (77.8%) as part of the initial diagnostic evaluation. CT was the most frequently utilized modality (n=5, 55.6%), followed by ultrasound (n=1, 11.1%). Three patients required additional advanced imaging, including MRI or positron emission tomography (PET), for further characterization or staging.

Eight of nine cases (88.9%) underwent initial needle biopsy prior to definitive diagnosis; one case (Deng *et al.*, 2021) proceeded directly to excisional biopsy without a prior needle biopsy attempt. Of the 8 cases with initial

Table 4 Individual patient characteristics and outcomes

Reference	Year	Age (years)	Primary HCC treatment	Metastasis site	Timing	AFP at metastasis	Diagnostic method	Salivary metastasis treatment	Survival (months)
Current case	2024	64	TACE → liver transplant	L parotid (masseter)	Metachronous (30 mo post-Tx)	Normal	CNB ×2 non-dx; surgical excision	Superficial parotidectomy + infratemporal resection; lenvatinib; SBRT	13[†]
Deng *et al.* (10)	2021	55	Multiple resections	R submandibular	Metachronous (96 mo)	NR	Excisional biopsy	Submandibular resection	6
Aiyer *et al.* (11)	2019	68	NR	R parotid + bone	Synchronous	Elevated	FNAB ×2 (2nd diagnostic)	Non-operative (chemo + RT)	NR
Elzouki *et al.* (12)	2014	66	NR	R parotid	Synchronous	Elevated	Surgical excision	Non-operative (supportive care)	2
Yu *et al.* (13)	2013	36	Right hepatectomy	L parotid	Metachronous (8 mo)	Normal	Surgical excision	Superficial parotidectomy	12[‡]
Moore *et al.* (14)	2010	75	NR	R parotid + bone + abdominal wall	Synchronous	NR	FNAB ×2 (2nd diagnostic)	Non-operative (sorafenib)	24[‡]
Vitale *et al.* (15)	2009	82	NR	R parotid	Synchronous	Normal	Surgical excision	Non-operative (supportive care)	6
Romanas *et al.* (16)	2004	47	NR	R parotid	Synchronous	NR	FNAB ×1 (diagnostic)	Non-operative (sorafenib + RT)	4
Dargent *et al.* (17)	1998	62	NR	L parotid	Synchronous	Elevated (3,520 ng/mL)	FNAB ×1 (diagnostic)	Non-operative (tamoxifen + RT)	NR

[†], died of disease; [‡], alive at last follow-up (censored). AFP, alpha-fetoprotein; CNB, core needle biopsy; dx, diagnostic; FNAB, fine-needle aspiration biopsy; HCC, hepatocellular carcinoma; L, left; mo, months; NR, not reported; R, right; RT, radiation therapy; SBRT, stereotactic body radiation therapy; TACE, transarterial chemoembolization; Tx, transplant.

needle biopsy, a diagnostic FNAB, defined as one yielding cytologic evidence sufficient to confirm the diagnosis of metastatic HCC without requiring further tissue sampling, was achieved on the first attempt in only 2 cases (25.0%): Romanas *et al.* [2004] and Dargent *et al.* [1998]. The remaining 6 cases (75.0%) had non-diagnostic initial biopsies, yielding benign or fibrofatty tissue, atypical cells, insufficient cellularity, or nonspecific findings. Among these, 2 cases (Aiyer *et al.*, 2019; Moore *et al.*, 2010) underwent repeat FNAB, which confirmed HCC metastasis on the second attempt. The remaining 4 cases required excisional or surgical biopsy for definitive diagnosis: Elzouki *et al.* [2014], Yu *et al.* [2013], Vitale *et al.* [2009], and our institutional case. The full diagnostic biopsy pathway across all 9 cases is summarized in *Table 4*.

Serum AFP levels (ng/mL) at the time of salivary metastasis diagnosis showed considerable heterogeneity. AFP was classified as elevated if reported above each study's stated upper limit of normal (ULN; typically ≤20 ng/mL) or if authors explicitly described it as elevated; normal if a value within the reference range was provided or the authors described it as within normal limits; and NR if no value or qualitative status was available. By these criteria, AFP was elevated in 3 patients (33.3%): Dargent *et al.* (3,520 ng/mL), Elzouki *et al.* (elevated, specific value NR), and Aiyer *et al.* (elevated, specific value NR). AFP was normal in 3 patients (33.3%): Yu *et al.*, Vitale *et al.*, and the current case. AFP was NR in the remaining 3 patients (33.3%): Romanas *et al.*, Moore *et al.*, and Deng *et al.* This distribution suggests that AFP may not reliably serve as a sensitive biomarker for identifying HCC salivary gland metastasis, as one-third of patients with confirmed metastasis had normal AFP levels in this series.

Immunohistochemical (IHC) and molecular profiles

IHC details were reported in 7 of 9 cases; Dargent *et al.* [1998] and Deng *et al.* [2021] did not include IHC data. Hepatocyte paraffin antigen-1 (HepPar-1) was the most frequently employed lineage marker, confirmed positive in 5 cases (Yu *et al.*, 2013; Moore *et al.*, 2010; Vitale *et al.*,

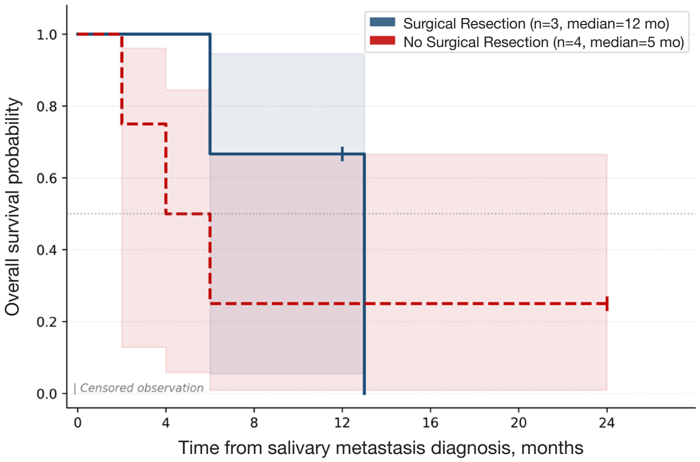

Figure 2 Kaplan-Meier survival analysis of patients with HCC metastasis to salivary glands (n=7 patients with survival data). HCC, hepatocellular carcinoma.

2009; Aiyer *et al.*, 2019; Elzouki *et al.*, 2014). Several cases supplemented HepPar-1 with additional hepatocellular markers: arginase-1 positivity was reported by Aiyer *et al.* [2019] and in our institutional case; AFP immunostaining was positive in Aiyer *et al.* [2019] and Romanas *et al.* [2004]; polyclonal carcinoembryonic antigen (CEA) with a canalicular staining pattern (characteristic of hepatocellular differentiation) was noted in Elzouki *et al.* [2014] and Romanas *et al.* [2004]. Vitale *et al.* [2009] additionally reported CD10 and CK8/18 positivity. Our institutional case employed a contemporary HCC panel [arginase-1+, Glypican-3+, programmed death-ligand 1 (PD-L1) 2–4%] without HepPar-1. Across all cases with available IHC data, no primary salivary gland markers were reported as positive, supporting a diagnosis of metastatic rather than primary salivary malignancy in every instance.

Formal molecular profiling was NR in any of the eight literature cases. Our institutional case underwent next-generation sequencing, demonstrating a tumor mutational burden (TMB) of 5.69 mutations/megabase (TMB-low) and microsatellite stable (MSS) status, consistent with the molecular landscape of conventional HCC and suggesting limited predicted response to immune checkpoint inhibition.

Treatment approaches

Treatment strategies varied based on the extent of disease, patient performance status, and institutional preferences. The time from salivary metastasis diagnosis to treatment initiation was explicitly reported in only 2 of 9 cases; for the remainder, timing was NR in the source publication. Three patients (33.3%) underwent surgical resection of the salivary metastasis, including superficial parotidectomy (n=2) and submandibular gland resection with reconstructive flap repair (n=1); Deng *et al.* reported surgery performed 1 week after admission, while the operative interval was NR for Yu *et al.* and Moore *et al.* All three achieved negative margins without major reported complications, and none received adjuvant systemic therapy. The remaining 6 patients (66.7%) were managed non-operatively. Of these, 4 patients (44.4% of the total cohort) received systemic therapy: sorafenib (Moore *et al.* and Romanas *et al.*), combination chemotherapy with nanoxel and carboplatin followed by concurrent chemoradiation (70 Gy in 35 fractions; Aiyer *et al.*), and hormonal therapy with tamoxifen plus local irradiation (Dargent *et al.*); treatment initiation timing was NR in all four. Palliative radiation alone was utilized in 1 patient (Romanas *et al.*) in the setting of progressive disease. The remaining 2 non-operative patients (22.2%), Elzouki *et al.* and Vitale *et al.*, received supportive care only, reflecting advanced disease burden and poor performance status.

Outcomes

Survival data were available for 7 of 9 patients (77.8%). Among these seven patients, the median overall survival from the time of diagnosis of salivary metastasis was 6 months (range, 2–24 months). Five deaths were documented during follow-up (71.4% mortality rate among patients with outcome data).

Kaplan-Meier survival curves stratified by treatment approach are presented in *Figure 2* for descriptive purposes.

Patients who underwent surgical resection of salivary metastasis (n=3) had a median survival of 12 months, compared with 5 months for those managed non-operatively (n=4 with survival data). This numerical difference should be interpreted with caution given the small sample size; no formal statistical comparison was performed.

Synthesis of results

Synthesizing findings across all 9 cases reveals a consistent clinical profile for HCC metastasis to the salivary glands. This rare entity demonstrates striking male predominance (100%), with no female cases identified despite a comprehensive systematic search. The parotid gland represents the overwhelmingly dominant anatomical site (88.9%), likely reflecting its unique embryological development with intraparenchymal lymphatic incorporation.

The clinical presentation pattern shows that two-thirds of cases present synchronously with primary HCC diagnosis, suggesting that salivary metastasis may represent a manifestation of advanced, disseminated disease at initial presentation. However, the occurrence of metachronous cases, including our transplant recipient with a 2.5-year disease-free interval, indicates that late metastasis can occur even after apparently successful primary HCC treatment.

Diagnostic challenges are evident in the poor performance of FNAB, which rarely provides a definitive diagnosis. This finding has important clinical implications, as it suggests that clinicians consider earlier surgical biopsy when HCC salivary metastasis is suspected, rather than relying on repeated FNABs that may delay diagnosis. The variable AFP elevation pattern further complicates diagnosis, as normal AFP does not exclude metastatic disease.

Treatment approaches varied substantially, reflecting the absence of standardized guidelines for this rare entity. While surgical resection demonstrated numerically superior median survival (12 versus 5 months), this observation is descriptive given the small sample size and should not be interpreted as a statistically or causally established difference. The overall poor prognosis is reflected in the 6-month median overall survival, consistent with outcomes reported for other sites of distant HCC metastasis.

The complete absence of female cases, despite HCC occurring in both sexes (though male-predominant), raises intriguing biological questions about sex-specific metastatic patterns that warrant further investigation.

Institutional case report

A 64-year-old male with alcoholic cirrhosis-related HCC underwent deceased donor liver transplantation in 2019. His extensive medical comorbidities included coronary artery disease, type 2 diabetes mellitus, alcohol use disorder, hyperlipidemia, hypertension, and tobacco use, reflecting the complex multimorbidity typical of transplant recipients.

HCC was initially diagnosed in September 2018 during routine cirrhosis surveillance, revealing two Liver Imaging Reporting and Data System (LI-RADS) 5 lesions (29 mm in segment 4A, 18 mm in segment 8) on contrast-enhanced MRI. He underwent bridging locoregional therapy with TACE while awaiting transplantation. In 2019, surgeons successfully performed a liver transplantation from a deceased donor, using standard immunosuppression that included tacrolimus and mycophenolate mofetil.

Post-transplant surveillance with serial cross-sectional imaging and AFP monitoring demonstrated no evidence of HCC recurrence for approximately 30 months. In May 2023, the patient presented with a left facial mass. CT maxillofacial imaging revealed a 3.8 cm enhancing lesion in the left masseter with associated zygomatic arch destruction. MRI of the face demonstrated a restricted diffusion lesion without regional nodal involvement or perineural spread.

The diagnostic workup was notable for two sequential non-diagnostic tissue biopsies. A first core needle biopsy performed in June 2023 yielded only benign fibrofatty tissue; a repeat core needle biopsy performed two weeks later was again non-diagnostic. Given the radiographic suspicion for malignancy and inability to establish a tissue diagnosis percutaneously, the case was reviewed at multidisciplinary tumor board and the patient proceeded to surgical resection. In July 2023, he underwent left superficial parotidectomy with infratemporal tumor resection. Intraoperative frozen section confirmed malignancy, and final histopathology demonstrated moderately to poorly differentiated metastatic HCC with negative surgical margins. Immunohistochemistry revealed arginase-1 positivity, glypican-3 positivity, and PD-L1 expression of 2–4%, consistent with hepatocellular origin. Primary salivary gland markers were negative. Molecular profiling demonstrated a TMB of 5.69 mutations/megabase (TMB-low) and MSS status. Serum AFP remained within normal limits throughout the diagnostic workup.

Disease progression was first evident in November 2023, approximately 4 months postoperatively, with local facial recurrence. The patient was started on lenvatinib (12 mg orally once daily) and completed stereotactic body radiation therapy (SBRT) to the facial recurrence site (30 Gy in 5 fractions) in late November 2023. By December 2023, imaging demonstrated new pulmonary nodules and a T9 vertebral lytic lesion. MRI of the spine in January 2024 revealed T5–T6 epidural extension with mass effect on the spinal cord; the patient was evaluated by neurosurgery but was deemed not a surgical candidate in February 2024. He completed palliative spinal radiation (30 Gy in 10 fractions) in March 2024. His clinical status deteriorated rapidly thereafter, with a fall admission in mid-March 2024 in the setting of cachexia and an Eastern Cooperative Oncology Group (ECOG) performance status of 3. He was enrolled in hospice in April 2024 and died on June 5, 2024, 13 months after the initial diagnosis of salivary metastasis. This case illustrates the clinical complexity of managing rare metastatic patterns in immunosuppressed transplant recipients and underscores that extended disease-free intervals do not preclude late distant metastasis.

Discussion

Key findings

This systematic review represents the first comprehensive analysis of HCC metastasis to salivary glands, identifying only nine documented cases worldwide despite HCC's substantial global disease burden. This extreme rarity underscores the clinical significance of recognizing and characterizing this unusual metastatic pattern.

Our analysis reveals several critical findings. First, all nine reported cases of HCC salivary gland metastasis occurred in males (100%), exceeding the typical 2–3:1 male predominance observed in general HCC populations; whether this reflects true biological sex-specificity or reporting artifact cannot be determined from this series. Second, the parotid gland is overwhelmingly the dominant anatomical site (88.9%), with submandibular involvement rare and no sublingual cases identified. Third, two-thirds of cases present synchronously with primary HCC diagnosis, though metachronous presentation can occur years later, including 2.5 years post-transplant in our institutional case. Fourth, fine-needle aspiration demonstrates significant limitations, with a diagnostic yield of only 25%, typically necessitating surgical biopsy for definitive diagnosis. Fifth, prognosis is consistently poor with a median survival of 6 months in this series; surgical resection demonstrated numerically superior outcomes (12 versus 5 months), though this observation is descriptive given the small sample size. Finally, the high prevalence of metabolic comorbidities, particularly alcohol use disorder (55.6%) and diabetes mellitus (33.3%), reflects typical HCC risk profiles and may influence treatment tolerability and outcomes.

Strengths

This systematic review has several important strengths. It represents the first systematic synthesis of evidence regarding HCC metastasis to salivary glands, addressing a previously uncharacterized clinical entity. We adhered to PRISMA 2020 guidelines with prospective protocol registration (PROSPERO CRD42024559907), a comprehensive search strategy across six major electronic databases with no language restrictions, and dual independent review of study selection and data extraction. The inclusion of our institutional transplant case provides the first reported example of this phenomenon in a liver transplant recipient, highlighting implications for post-transplant surveillance. All included cases had pathological confirmation of both primary HCC and salivary metastasis, ensuring diagnostic certainty. Patient-level data extraction enabled detailed characterization of clinical features, diagnostic approaches, treatment strategies, and outcomes that would not be possible with study-level synthesis alone.

Limitations

This study has several important limitations that must be considered when interpreting the findings. First, the sample size is extremely small (n=9 cases), which severely limits statistical power and precludes identification of definitive prognostic factors or reliable subgroup comparisons; the survival analysis should be regarded as descriptive only. Second, the evidence base consists exclusively of case reports, representing the lowest level of evidence in the hierarchy; although this reflects the extreme rarity of the condition, it introduces inherent risk of selection and reporting bias. Third, survival data were unavailable for 2 of 9 patients (Dargent *et al.*, 1998; Aiyer *et al.*, 2019), limiting the completeness of survival estimates. Fourth, the time from salivary metastasis diagnosis to treatment initiation was explicitly reported in only 1 of 9 cases, precluding any analysis of treatment delay and outcomes. Fifth, substantial heterogeneity exists in reporting completeness across included studies: AFP values were NR in 3 patients (33.3%), HCC staging at initial diagnosis was inconsistently documented, and follow-up protocols varied widely. Sixth, publication bias likely favors cases with unusual or dramatic presentations, potentially overrepresenting certain clinical features and distorting the apparent clinical profile of this entity. Seventh, we cannot exclude the possibility that some cases were undiagnosed or misclassified as primary salivary malignancies, particularly when the patient's HCC history was unknown to head and neck surgeons. Eighth, no quality-of-life or functional outcome data were reported in any included case. Finally, the geographic distribution skews toward high-income countries with advanced healthcare infrastructure, which may limit generalizability to resource-limited settings where HCC is most prevalent.

Comparison with similar research

To our knowledge, no previous systematic review has specifically examined HCC metastasis to salivary glands. However, we can contextualize our findings within the broader literature on HCC extrahepatic metastasis and salivary gland metastases from distant primaries.

Natsuizaka *et al.* reported that extrahepatic HCC metastasis occurs in 13.5–42% of patients, with the lungs (55%), lymph nodes (53%), and bone (28%) as the most common sites (4). Salivary glands are not separately enumerated in large HCC metastasis series, consistent with the extreme rarity documented in this series. The median survival of 6 months in our cohort aligns closely with reported outcomes for other distant HCC metastases, suggesting that prognosis is determined more by the presence of distant metastasis than by specific anatomical location.

Batsakis *et al.* reported that metastases account for less than 10% of all salivary gland malignancies, with most originating from head and neck primaries (5). Our finding of parotid predominance (88.9%) parallels general patterns in salivary metastases, where parotid involvement occurs in 80–85% of cases regardless of primary site. This consistency across different primary tumor types suggests that the parotid's unique lymphatic architecture, rather than HCC-specific tropism, may drive anatomical predilection.

The diagnostic challenge we identified with fine needle aspiration (25% initial diagnostic yield) is consistent with Rohra *et al.*'s analysis of salivary metastases, which found FNAB sensitivity of only 48% for metastatic disease compared to 96% for primary salivary malignancies (18). Notably, 2 of the 6 cases with non-diagnostic initial FNABs (Ayer *et al.*, 2019; Moore *et al.*, 2010) achieved a definitive diagnosis on repeat FNAB, suggesting that a second aspiration attempt may be warranted before proceeding to surgical biopsy in select cases. Nevertheless, 4 of 8 cases ultimately required excisional or surgical biopsy, and this overall limitation of aspiration cytology appears generalizable across metastatic primaries, not specific to HCC.

Notably, all nine cases in our cohort occurred in males (100%), exceeding patterns reported for other HCC metastatic sites, where male predominance typically mirrors general HCC demographics (2–3:1 ratio) (19). This may suggest sex-specific biological factors in metastatic tropism to salivary glands that warrant investigation, though the small sample size precludes any firm conclusions.

Our transplant case represents a novel contribution, as no previous reports of HCC salivary metastasis in liver transplant recipients exist. Among the three metachronous cases in our cohort, the median disease-free interval from primary HCC diagnosis to salivary metastasis was 2.5 years (range, 8 months to 8 years), reflecting our own data rather than an externally derived estimate. This is consistent with well-described patterns of late HCC recurrence following liver transplantation, in which recurrences beyond 2 years, and even beyond 5 years, post-transplant have been reported (20-23).

Across the cohort, IHC profiles consistently reflected hepatocellular lineage without evidence of primary salivary differentiation. HepPar-1 was the predominant marker used in the literature cases, though its moderate specificity for hepatocellular origin is well recognized (24). More specific contemporary markers, arginase-1 and glypican-3, were employed in only a subset of cases, most notably in Aiyer *et al.* [2019] and in our institutional case. The absence of formal molecular profiling across all eight literature cases precludes conclusions regarding HCC molecular subtype and salivary tropism. Our institutional case demonstrated TMB-low and MSI-stable status, consistent with the predominant molecular landscape of conventional HCC; however, whether specific molecular subtypes are overrepresented in cases with salivary metastasis cannot be determined from the current evidence base.

Explanations of findings

Biological and anatomical mechanisms

The mechanisms by which HCC reaches the salivary glands remain incompletely understood, but several converging biological and anatomical factors likely contribute.

Three principal routes of metastatic spread are plausible: hematogenous dissemination, lymphatic permeation, and retrograde venous flow via Batson's vertebral venous plexus (25). HCC is among the most hypervascular of solid tumors, with extensive arteriovenous shunting that facilitates the shedding of circulating tumor cells into the systemic circulation. Vascular invasion, both macroscopic and microscopic, is a hallmark of advanced HCC and correlates strongly with extrahepatic spread (26). Once in the systemic circulation, HCC-derived tumor cells may arrest in the microvasculature of distant organs, including the head and neck.

The striking predilection for the parotid gland (88.9% of cases) reflects its unique embryological and anatomical properties. Unlike the submandibular and sublingual glands, which fully encapsulate prior to lymphatic tissue maturation, the parotid undergoes delayed capsule formation, allowing intraparenchymal lymph nodes to become incorporated within its substance (7). The parotid is thus the only major salivary gland with regular intraglandular lymph nodes, making it a lymphatic crossroads that receives drainage from the scalp, periauricular skin, external auditory canal, and conjunctiva. This architecture provides both a lymphatic and a hematogenous nidus for metastatic seeding that is structurally absent in the submandibular and sublingual glands. The parotid's rich arterial supply and the presence of specialized arteriovenous communications may further facilitate tumor cell arrest and extravasation (27).

The absolute male predominance observed in our cohort (100%) merits biological consideration beyond the known 2–3:1 male-to-female ratio of HCC itself (19). Androgen receptor (AR) signaling plays a well-established oncogenic role in HCC, promoting tumor growth and invasion (28). Whether AR is expressed in parotid tissue at levels sufficient to influence metastatic tropism remains an open question, but androgen-driven modulation of the tumor microenvironment and lymphatic function represents a biologically plausible mechanism. Sex-specific differences in tumor microenvironment, immune surveillance, and lymphatic function may further contribute. These hypotheses are speculative given the small sample size, but they represent a mechanistically plausible basis for the observed sex disparity that warrants dedicated investigation.

Finally, immunosuppression in the context of liver transplantation represents a distinct biological variable. Calcineurin inhibitors such as tacrolimus promote tumor growth and metastatic potential through direct effects on transforming growth factor-beta (TGF-β) signaling and impairment of natural killer cell surveillance (29). The 2.5-year disease-free interval followed by multifocal metastatic recurrence in our institutional case is consistent with immunosuppression-mediated uncoupling of dormant micrometastatic disease, a phenomenon increasingly recognized in post-transplant oncology (20,21). The interplay between HCC molecular biology, donor-recipient immune dynamics, and immunosuppressive regimen intensity likely modulates both the timing and pattern of late metastatic recurrence in this population.

Clinical implications

The findings of this review carry several practical implications for gastrointestinal (GI) oncologists and hepatobiliary clinicians managing HCC patients. Most directly, the poor diagnostic yield of FNAB (25% on first attempt) argues against a strategy of repeated needle aspiration when salivary metastasis is clinically suspected. Given that four of eight cases in this series ultimately required excisional or surgical biopsy for definitive diagnosis, a low threshold for early surgical referral is reasonable when initial tissue sampling is non-diagnostic and clinical suspicion remains high. Importantly, normal AFP does not exclude salivary metastasis; one-third of confirmed cases had AFP within normal limits, so a negative AFP should not reassure the clinician or delay tissue sampling.

Several clinical features from this series may serve as practical prompts for heightened suspicion. A new, painless parotid or submandibular mass in a male HCC patient, whether at initial diagnosis or at any point during follow-up, warrants prompt evaluation and should not be attributed to a benign parotid lesion without pathological confirmation. The risk appears particularly concentrated in male patients with metabolic comorbidities (alcohol use disorder, diabetes mellitus, cirrhosis), who represent the majority of cases in this series. The synchronous presentation rate of 66.7% means that comprehensive metastatic staging at HCC diagnosis should include clinical assessment of the head and neck region, even in the absence of specific symptoms. For metachronous presentations, the interval can span years: our transplant recipient developed salivary metastasis 2.5 years after transplantation, and one literature case occurred 8 years after initial HCC diagnosis. This argues for sustained, symptom-driven vigilance rather than a fixed surveillance window. When a salivary metastasis is identified, concurrent extrahepatic disease at other sites, particularly bone identified in 22.2% of cases, should be excluded, and cross-sectional staging with PET/CT may be warranted given the systemic implications.

Treatment decisions in this context are inherently individualized. While surgical resection was associated with numerically longer survival in this small series, the overall prognosis remains poor regardless of approach (median 6 months overall), and the decision to operate should account for the extent of systemic disease, patient performance status, and goals of care. A multidisciplinary approach involving hepatology, surgical oncology, otolaryngology, medical oncology, and palliative care is appropriate for these complex cases.

Implications and actions needed

Clinical practice recommendations

Taken together, the evidence from this series supports several practice-oriented recommendations for clinicians managing HCC patients. First, HCC salivary gland metastasis should be included in the differential diagnosis of any new neck or facial mass in a male HCC patient, regardless of AFP status or time since initial diagnosis. Second, when FNAB is non-diagnostic in this clinical context, early surgical biopsy is preferable to repeated aspiration attempts. Third, identification of a salivary metastasis should trigger comprehensive systemic staging, given the frequency of concurrent extrahepatic disease. Fourth, treatment planning is best undertaken through a multidisciplinary framework, with otolaryngology and head and neck surgery involved early to facilitate surgical planning when operative intervention is being considered. Finally, prognosis should be communicated transparently: median survival in this series was approximately 6 months, and early integration of palliative care is appropriate for patients with advanced disease burden or limited performance status.

Surveillance protocol considerations for transplant recipients

Our institution reported the first case of metastatic HCC to a salivary gland in a liver transplant recipient, highlighting gaps in current post-transplant surveillance. Standard protocols typically monitor for 2 years, but our case, occurring 2.5 years post-transplant, suggests extended surveillance is needed for high-risk patients. Routine head and neck exams may not be cost-effective, but clinicians should specifically ask about new neck symptoms during follow-up and educate patients on warning signs. Prompt imaging should be pursued when symptoms arise, as parotid metastases can mimic benign conditions. More intensive surveillance may be justified in male recipients with metabolic syndrome or alcohol-related HCC, though criteria need further study. Clear communication of HCC history to specialists is essential. This case illustrates the need for tailored, symptom-driven surveillance and better multidisciplinary collaboration to improve outcomes.

Research priorities

Further research is needed to clarify why HCC occasionally spreads to salivary glands. Mechanistic studies should examine sex-specific and molecular factors influencing metastatic patterns. Building multi-institutional registries will allow for more robust analysis of prognostic factors and treatment outcomes in this rare setting. Broader analyses of head and neck metastases could reveal common biological mechanisms, and molecular profiling of tumor clones may help map metastatic progression. Enhanced surveillance protocols, including feasibility of head-and-neck exams for high-risk transplant patients, should be piloted. Future case reports should address quality-of-life and functional outcomes. Finally, registry studies in transplant recipients may clarify how immunosuppression alters disease course. As HCC outcomes improve, investigating late and atypical metastases will be essential for optimal long-term care.

Conclusions

HCC metastasis to salivary glands represents an exceedingly rare clinical entity observed exclusively in males across all reported cases, with consistently poor prognosis in this series. The parotid gland's predominance (88.9%) likely reflects its unique intraparenchymal lymphatic architecture, arising from delayed embryological encapsulation, which creates multiple convergent pathways for metastatic seeding that are structurally absent in other major salivary glands. Fine needle aspiration demonstrated significant diagnostic limitations in this series, with surgical biopsy required in most cases for definitive pathological confirmation. Two-thirds of cases present synchronously with primary HCC diagnosis, though metachronous presentation can occur years after initial treatment, including post-transplantation. While surgical resection demonstrates numerically superior survival outcomes, the overall poor prognosis (median 6 months) mandates careful patient selection and individualized treatment decisions, balancing potential benefits against treatment-related morbidity. For liver transplant recipients, this rare phenomenon underscores the importance of extended surveillance beyond conventional sites and timeframes, with heightened awareness for unusual metastatic patterns as transplant survival continues improving. Future research through multi-institutional registries and mechanistic studies is essential to better characterize optimal diagnostic approaches, treatment strategies, and surveillance protocols for this challenging clinical entity.

Acknowledgments

The authors would like to thank the medical librarian at Geisinger Medical Center for assistance in developing the comprehensive search strategy across multiple databases. We acknowledge the Geisinger Health System's Department of Liver and Kidney Transplantation and the Geisinger Northeast General Surgery Residency for their institutional support of this research. We are grateful to the patients and families for their willingness to contribute to medical knowledge. Finally, we acknowledge all authors of the original case reports included in this systematic review, whose documentation enabled this synthesis of evidence.

Footnote

Reporting Checklist: The authors have completed the PRISMA reporting checklist. Available at https://jgo.amegroups.com/article/view/10.21037/jgo-2026-1-0015/rc

Peer Review File: Available at https://jgo.amegroups.com/article/view/10.21037/jgo-2026-1-0015/prf

Funding: None.

Conflicts of Interest: All authors have completed the ICMJE uniform disclosure form (available at https://jgo.amegroups.com/article/view/10.21037/jgo-2026-1-0015/coif). The authors have no conflicts of interest to declare.

Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. As the systematic review involved only published literature and did not include any human subjects research, it was determined to be exempt from additional ethics review under Geisinger Health System IRB guidelines.

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Cite this article as: Morada AO, Chen CK, Akateh C, Marvin MR, Kotru A. Hepatocellular carcinoma metastasis to salivary glands: a systematic review of diagnostic challenges and survival outcomes. J Gastrointest Oncol 2026;17(3):175. doi: 10.21037/jgo-2026-1-0015

